

ASHWIN SENTHIL ARUMUGAM

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SKILLS

Areas of Interest Full Stack Dev, Systems programming, Distributed Systems, Mobile Apps, Security, AI
Languages and Frameworks Java, C, Go, Python, Kotlin, C++, SQL, Android, ReactJS, NextJS, TailWind, Rest Api
Tools and Technologies AWS, Docker, Kubernetes, Linux, Apache, Jenkins, Tensorflow, PyTorch, Deep Learning

EXPERIENCE

SAP *Bengaluru, India*

Associate Developer | Tech : Java, Python, ABAP, API development, BOPF, UI5, JS, Fiori, ML Jul 2020 - Jul 2023

- Developed and optimized front end and back end modules in SAP Public Cloud solutions (SAP TM), leading to quicker calculation of freight transportation charges and efficient maintenance and display of agreements.
- Led the development of OData and Soap API to enable efficient transfer of transportation costs from business provider's system to SAP Cloud system.
- Revamped the system and database architecture of a legacy module in SAP TM to make it 90 percent faster to determine costs for transportation quotations.
- Built an Machine Learning model with 80 percent accuracy to extract key information from emails and text to automatically populate enterprise applications in SaaS solutions.

Samsung *Bengaluru, India*

Intern | Tech : Python, Deep Learning, Tensorflow, PyTorch, Machine Learning May 2019 - Jul 2019

- Assessed the capability of page ranking algorithms in identifying customer specific relevant search results and their ability to constantly reinforce over time.
- Constructed a Deep Learning ranking algorithm trained on millions of data points to efficiently rank apps in Samsung App Store. The model works with incorrect or incomplete queries from user as well to predict expected applications.

RESEARCH/OPEN SOURCE PROJECTS

Microsoft Restler Fuzzer Adapted Microsoft RESTler Fuzzer algorithm in Python to utilize the corpus data of successful api requests as seed for random value generation for new sequences. This approach helped find new vulnerabilities in 5G network components. ([GitHub](#))

CEG Prints | Android, Java, Python Flask, Firebase Engineered an android app and a Flask server to enable wireless printing from mobile devices in local stores. Published in PlayStore resulting in over **4k** downloads, this project received an award from tech forum club in Anna University. ([GitHub](#))

Key-Value Store using Paxos Implemented a distributed Paxos-protocol key-value store in GoLang, ensuring data consistency and fault tolerance for the Distributed Systems course. ([GitHub](#))

AI image classification Using Transfer Learning to train Vision Transformer model on custom dataset, the model classified images as Real or AI generated with 80 percent accuracy ([GitHub](#))

Aditi - Accessible web application Developed an inclusive web application in partnership with an NGO, for users with physical impairments. Utilizing just the left mouse click, the user can navigate the entire application.

EDUCATION

Master of Science in Computer Science and Engineering *Expected May 2025*
Pennsylvania State University | University Park, PA *GPA: 3.78/4*

Bachelor of Technology in Information Technology *2016 - 2020*
College of Engineering Guindy, Anna University | Chennai, India *GPA: 8.43/10*